A PROPER DENSE OPEN IN A NOETHERIAN SPACE EXTENDS TO A DISTINGUISHED SQUARE WITH DENSE INTERSECTION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3757-3768 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 3770-3796, the complete author proof (25 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 20 of queue_spaces-pushouts.json (pb=1399, stmt 3757-3768, proof 3770-3796).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07ST: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3796–3796. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BA8: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3796–3796. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZV: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3796–3796. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06NH: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3796–3796. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Let $U \subset X$ be a proper dense open subspace. Then there exists an affine scheme V and an étale morphism $V \to X$ such that*

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *the open subspace* $W = U \cup \mathrm{Im}(V \to X)$ *is strictly larger than* U,
(2) $(U \subset W, V \to W)$ *is a distinguished square, and*
(3) $U \times_W V \to U$ *has dense image.*

Proof. Choose a stratification

$$\emptyset = U_{n+1} \subset U_n \subset U_{n-1} \subset \ldots \subset U_1 = X$$

and morphisms $f_p : V_p \to U_p$ as in Decent Spaces, Lemma 07ST. Let p be the smallest integer such that $U_p \not\subset U$ (this is possible as $U \neq X$). Choose an affine open $V \subset V_p$ such that the étale morphism $f_p|_V : V \to X$ does not factor through U. Consider the open $W = U \cup \mathrm{Im}(V \to X)$ and the reduced closed subspace $Z \subset W$ with $|Z| = |W| \setminus |U|$. Then $f^{-1}Z \to Z$ is an isomorphism because we have the corresponding property for the morphism f_p, see the lemma cited above. Thus $(U \subset W, f : V \to W)$ is a distinguished square. It may not be true that the open $I = \mathrm{Im}(U \times_W V \to U)$ is dense in U. The algebraic space $U' \subset U$ whose underlying set is $|U| \setminus \overline{|I|}$ is Noetherian and hence we can find a dense open subscheme $U'' \subset U'$, see for example Properties of Spaces, Proposition 06NH. Then we can find a dense open affine $U''' \subset U''$, see Properties, Lemmas 0BA8 and 01ZV. After we replace f by $V \amalg U''' \to X$ everything is clear. □